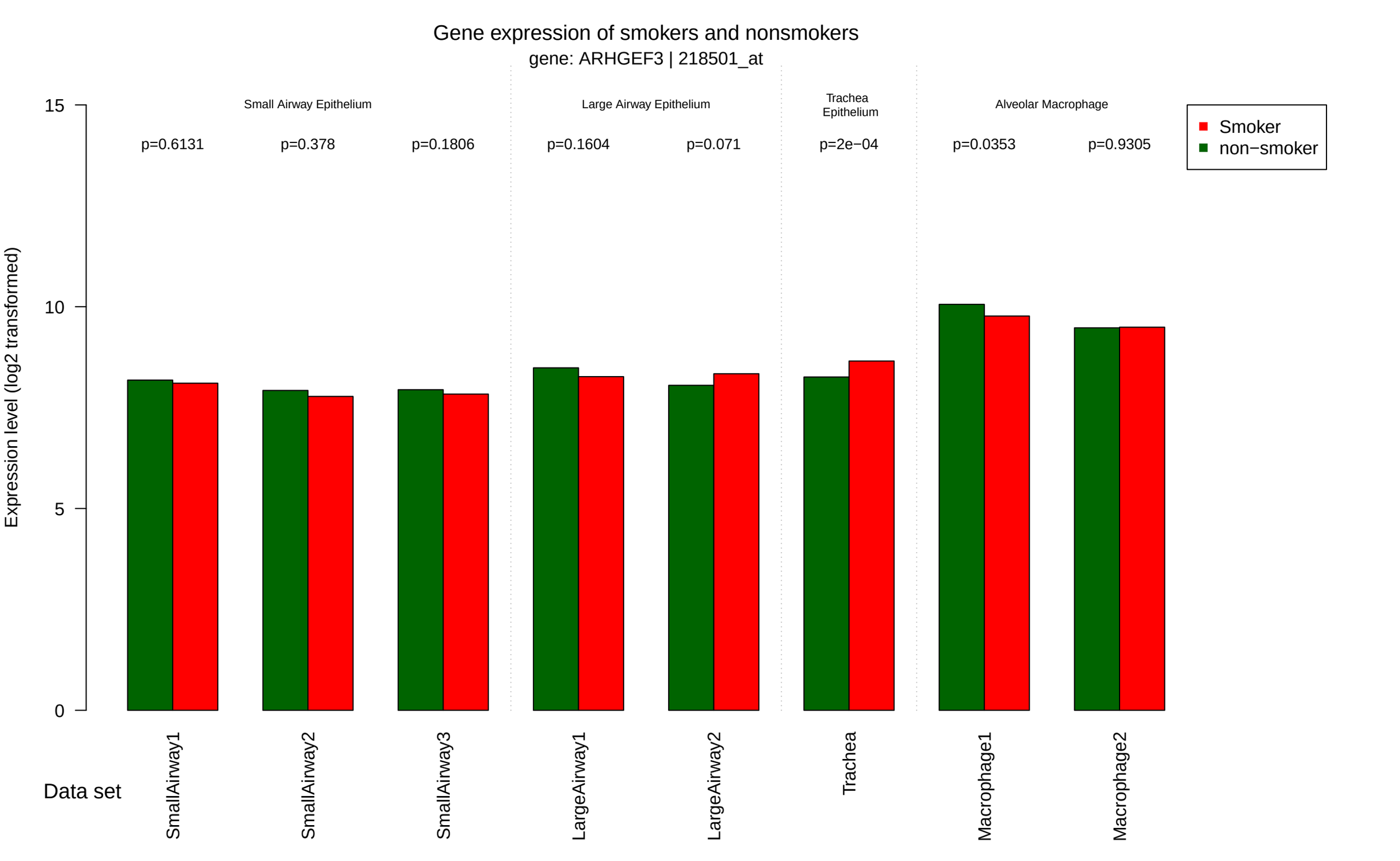
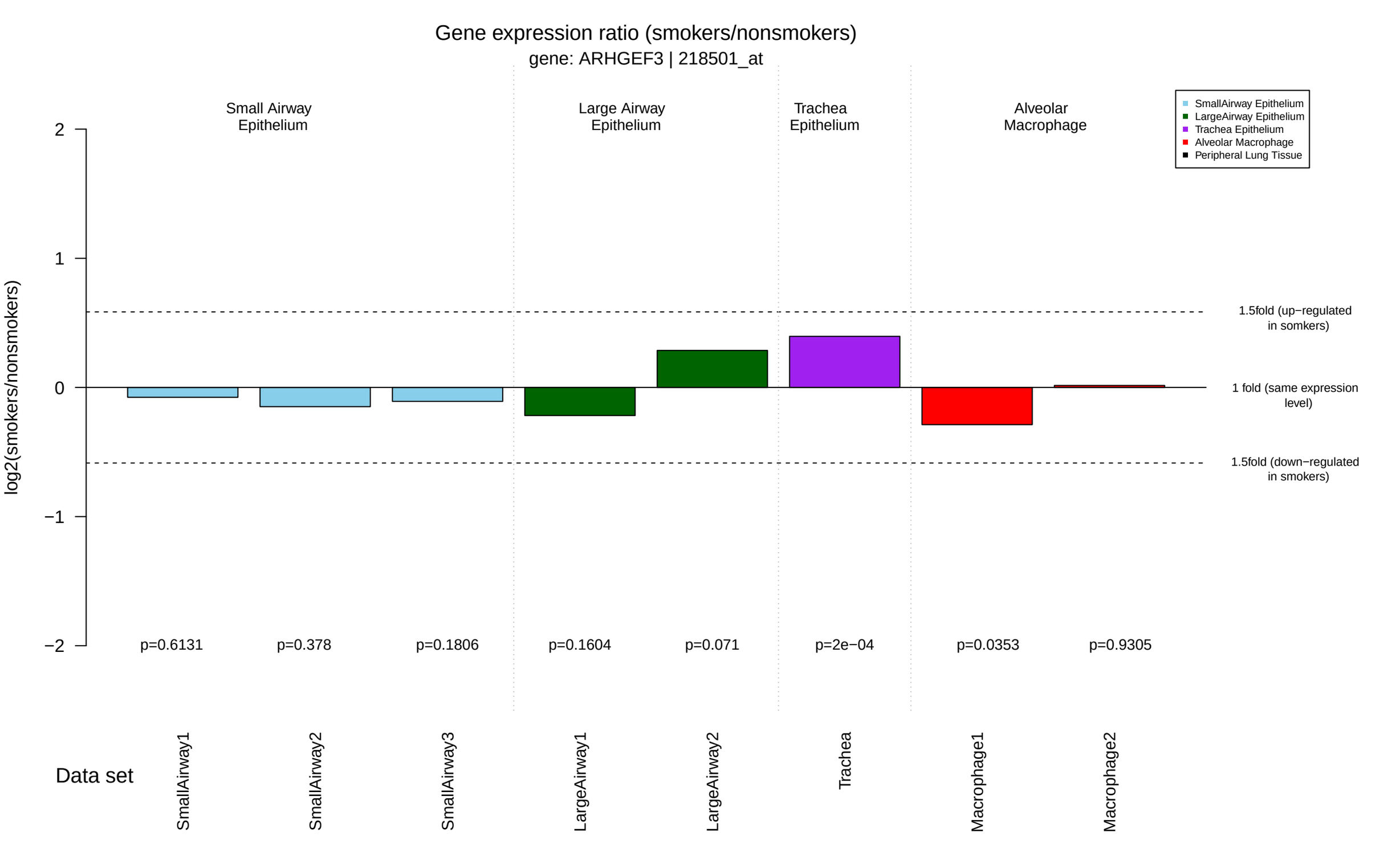
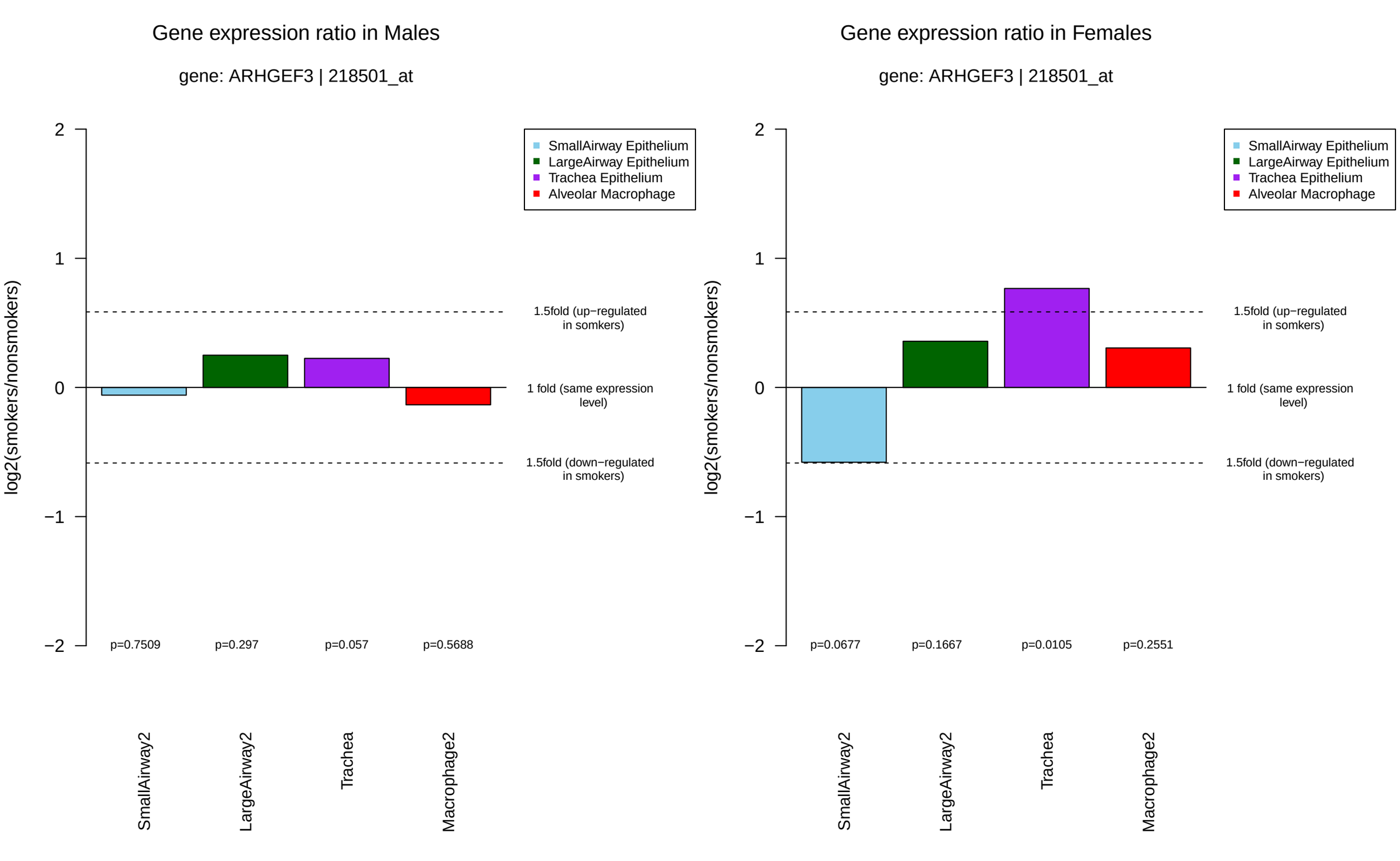
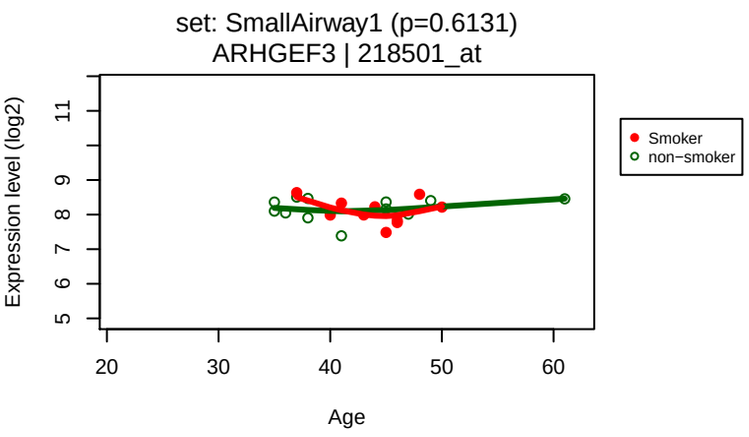
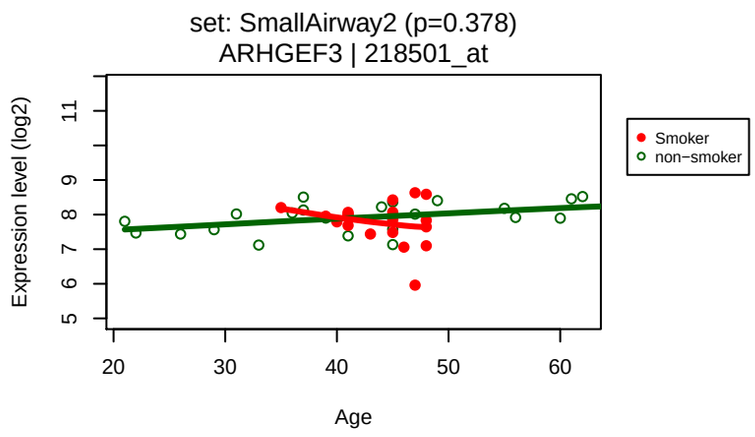
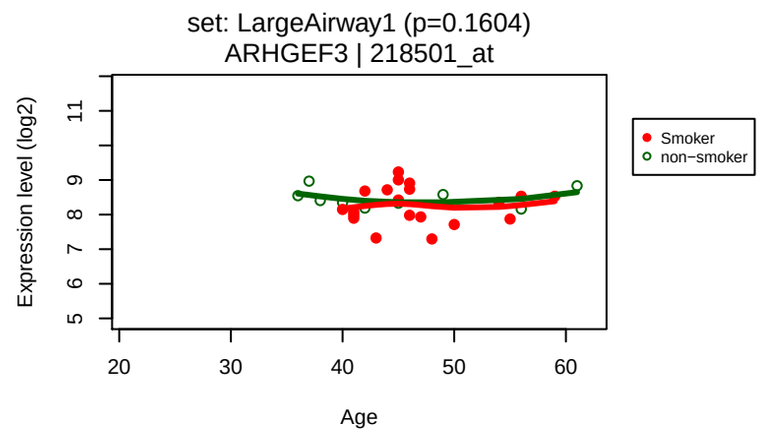

Gene expression of smokers and nonsmokers
gene: ARHGEF3 | 218501_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6131
p=0.378
p=0.1806
p=0.1604
p=0.071
p=2e−04
p=0.0353
p=0.9305
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

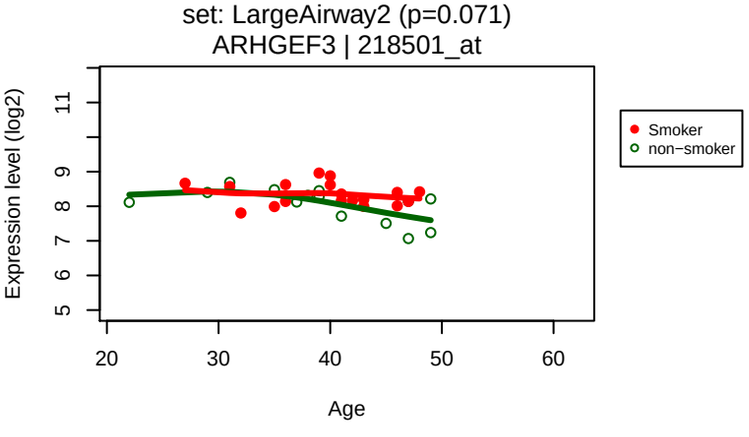

Gene expression ratio (smokers/nonsmokers)
gene: ARHGEF3 | 218501_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6131
p=0.378
p=0.1806
p=0.1604
p=0.071
p=2e−04
p=0.0353
p=0.9305
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

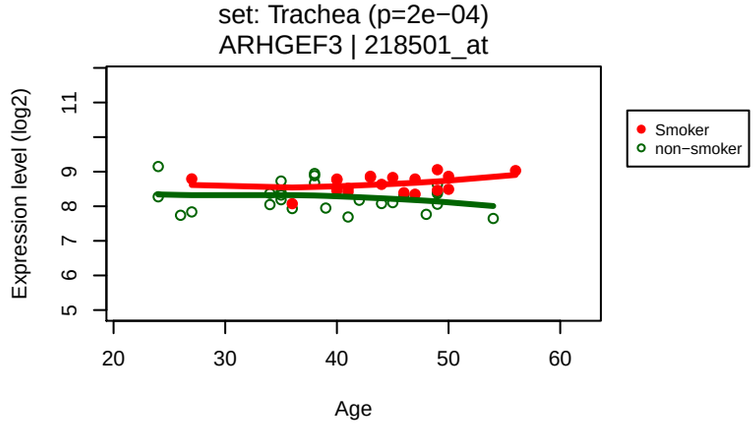

Gene expression ratio in Males
gene: ARHGEF3 | 218501_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7509
p=0.297
p=0.057
p=0.5688
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARHGEF3 | 218501_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0677
p=0.1667
p=0.0105
p=0.2551
SmallAirway2
LargeAirway2
Trachea
Macrophage2

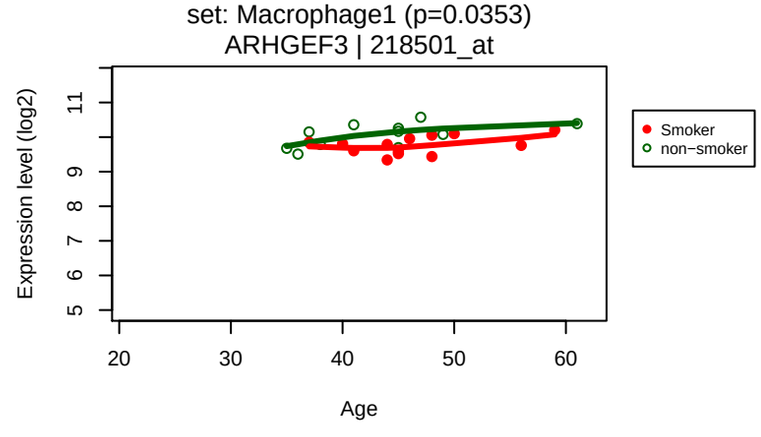

set: SmallAirway1 (p=0.6131)
ARHGEF3 | 218501_at
Expression level (log2)
Age
Smoker
non−smoker

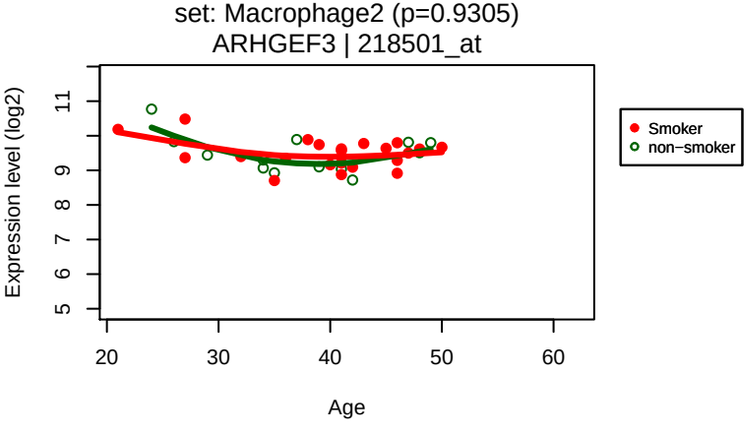

set: SmallAirway2 (p=0.378)
ARHGEF3 | 218501_at
Expression level (log2)
Age
Smoker
non−smoker

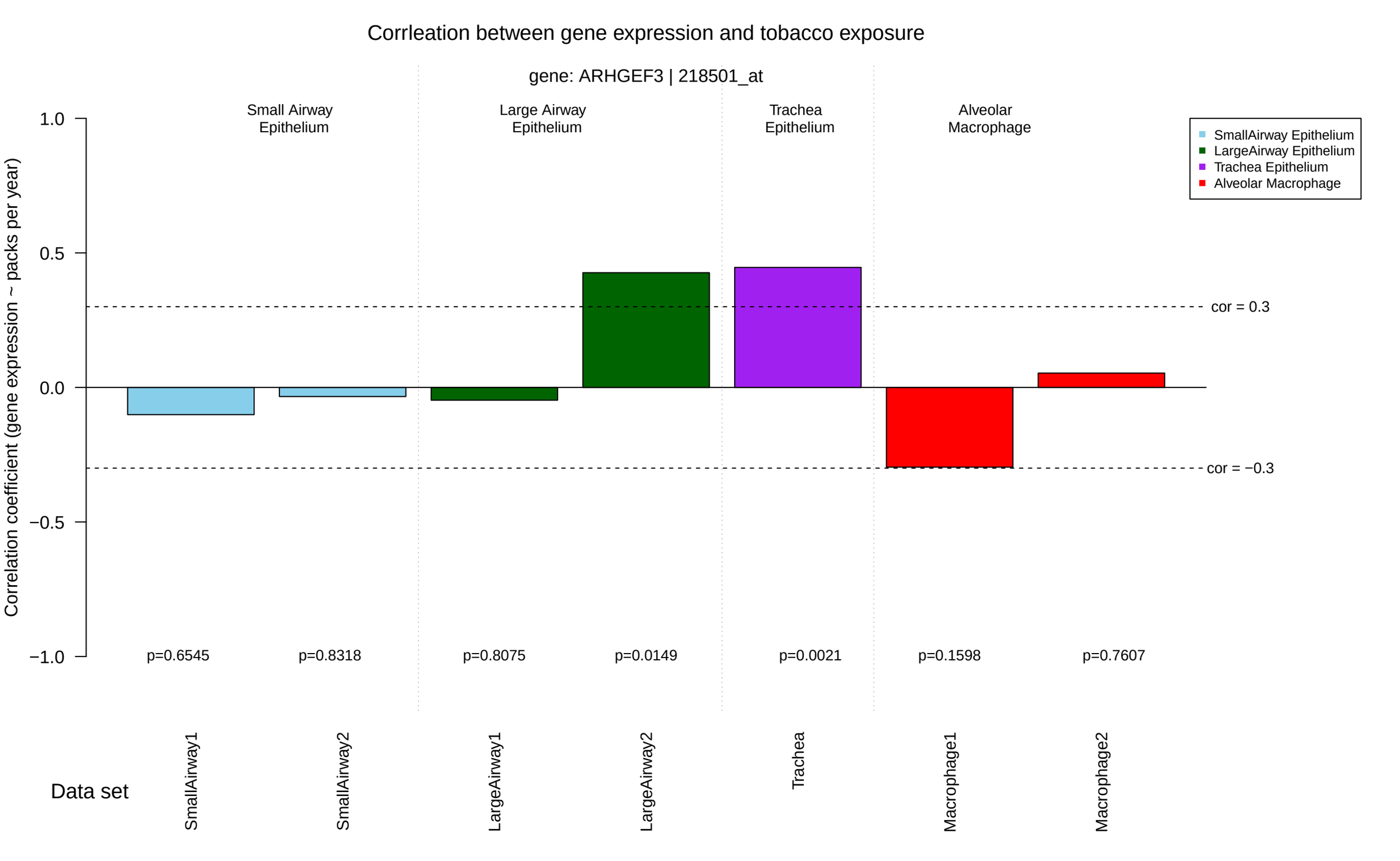

set: LargeAirway1 (p=0.1604)
ARHGEF3 | 218501_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.071)
ARHGEF3 | 218501_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=2e−04)
ARHGEF3 | 218501_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0353)
ARHGEF3 | 218501_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9305)
ARHGEF3 | 218501_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARHGEF3 | 218501_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6545
p=0.8318
p=0.8075
p=0.0149
p=0.0021
p=0.1598
p=0.7607
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2